Four convergence rates, computed as transfer–matrix contractions

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Abstract

A great many "rates" in analysis, dynamics and combinatorics are the exponential growth rate of a sum over words, $Z_L = \sum_{|I|=L} \omega(I)$. Whenever the weight ω is carried by a *finite transfer matrix* T – so that $Z_L = \langle u, T^L v \rangle$ – the rate is simply the top eigenvalue of T, and the *speed* at which the naive ratio Z_{L+1}/Z_L settles is the *spectral gap*. Below are four worked examples that make this concrete, from the trivial to the genuinely hard. Each states the object in one line, writes the series and its transfer engine explicitly, and reports the number we actually computed (code in `code/`). The reader who wants only the punchlines can read Table 1 and skip to the figures.

The one fact behind all four

Fix a finite alphabet, let $\mathcal{W}_L$ be the admissible words of length L, and let $\omega \geq 0$ be multiplicative along a word. The object of interest is

$$Z_L = \sum_{I \in \mathcal{W}_L} \omega(I), \qquad \lambda = \lim_{L\to\infty} Z_L^{1/L} \quad (\text{the } \textit{rate}), \qquad P = \log \lambda. \tag{1}$$

Say the family is *transfer-decomposable* if there is a fixed matrix $T \geq 0$ and boundary vectors u, v with

$$\boxed{Z_L = \langle u,\, T^L\, v\rangle \quad \text{for all } L.} \tag{2}$$

By Perron–Frobenius, if T is primitive with eigenvalues $|\lambda_1| > |\lambda_2| \geq \cdots$, then

$$\boxed{\lambda = \lambda_1 = \rho(T), \qquad \frac{Z_{L+1}}{Z_L} = \lambda_1\Big(1 + O(\delta^L)\Big), \qquad \delta := \frac{|\lambda_2|}{\lambda_1} \in [0,1).} \tag{3}$$

So the rate is an *eigenvalue* (computed once, exactly), and the "divide out the conjectured rate and watch the ratio go to 1" heuristic is not only correct – its convergence slope on a log plot *is* the spectral gap δ. The only question is whether a *finite* T exists:

Decomposable (finite T exists)	**Entangled** (it does not)
rate $= \rho(T)$, computed exactly; cost grows like the matrix dimension	no finite T; brute-force the q^L words and *fit* the rate
Examples 1–3	Example 4

	Quantity (known value)	Series & engine	What we computed
1	golden ratio φ = 1.61803398875	binary words, no "11"; $T = \left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$	ratio $\to \varphi$; error slope = gap φ^{-2} (meas. 0.381965)
2	hard-square constant $\kappa = 1.50304808$	strip width W; $\dim T_W = F_{W+2}$	κ to **13 digits** at $W = 14$
3	$\dim_H E_2$ = 0.5312805062772	Gauss-map operator $\mathcal{L}_s$	**15 digits** at grid $N = 16$; ratio test $\to 1$ iff $s = s^*$
4	matrix-product rate / affinity dim.	$T_I = T_{i_1} \cdots T_{i_L}$; graph K_L (no finite T)	surrogate = $\log \rho(T_1+T_2)$; affinity dim $s^* \approx 1.089$

Table 1: The four examples at a glance.

1 Golden-mean shift — the cleanest case

Compute. The number of binary strings of length L with no two adjacent 1s. The growth rate is the golden ratio.

Series. With $\mathcal{W}_L = \{x \in \{0,1\}^L : x_i x_{i+1} = 0 \ \forall i\}$ and $\omega \equiv 1$,

$$Z_L = \#\mathcal{W}_L = F_{L+2} \quad \text{(Fibonacci)}: \quad 3, 5, 8, 13, 21, 34, \ldots$$

Engine & rate. A word is built one bit at a time, the only memory being the previous bit, so $Z_L = \mathbf{1}^\top T^L \mathbf{1}$ with the 2×2 transfer matrix

$$T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}, \qquad \boxed{\lambda = \lambda_1 = \varphi = \tfrac{1+\sqrt{5}}{2}, \quad \delta = \frac{|\lambda_2|}{\lambda_1} = \varphi^{-2}.}$$

Result. $Z_{L+1}/Z_L \to \varphi = 1.6180339887\ldots$, and the error $|Z_{L+1}/Z_L - \varphi|$ falls geometrically with measured per-step factor 0.381965 against the predicted gap $\varphi^{-2} = 0.381966$ (Figure 1). *The slope of the error plot is the spectral gap* – equation (3) in its purest form.

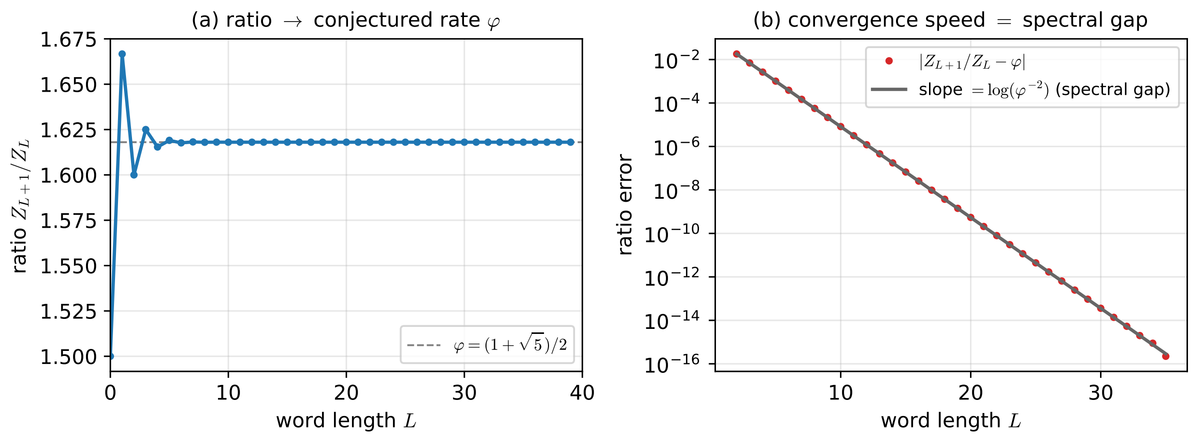

Figure 1: Golden-mean shift. (a) the ratio reaches φ; (b) the divided-out error decays at exactly the spectral-gap rate φ^{-2}.

2 Hard squares on a strip — width is the cost/accuracy dial

Compute. Ways to occupy cells of a $W \times L$ grid with no two occupied cells edge-adjacent (the hard-core lattice gas). The per-cell growth rate converges, as $W \to \infty$, to the hard-square constant $\kappa = 1.5030480824753\ldots$[1]

Series. $Z_{W,L}$ = number of independent sets of the $W \times L$ grid graph, summed along the length L.

Engine & rate. Read the grid column by column: the state is the occupied set of one width-W column, which must itself be an independent set of the path P_W. This gives a transfer matrix T_W with

$$\dim T_W = \#\{\text{independent sets of } P_W\} = F_{W+2} \sim \varphi^W, \qquad Z_{W,L} = \mathbf{1}^\top T_W^L \,\mathbf{1},$$

$$\boxed{\kappa = \lim_{W\to\infty} \rho(T_W)^{1/W} = \lim_{W\to\infty} \frac{\rho(T_W)}{\rho(T_{W-1})}.}$$

The matrix *dimension* $F_{W+2} \sim \varphi^W$ grows exponentially in the strip width W — so W is a dial that buys accuracy at exponential cost.

Result. The incremental ratio $\rho(T_W)/\rho(T_{W-1})$ converges *geometrically* and already matches κ to **13 digits** at $W = 14$ ($1.503048082\ldots$); see Figure 2.

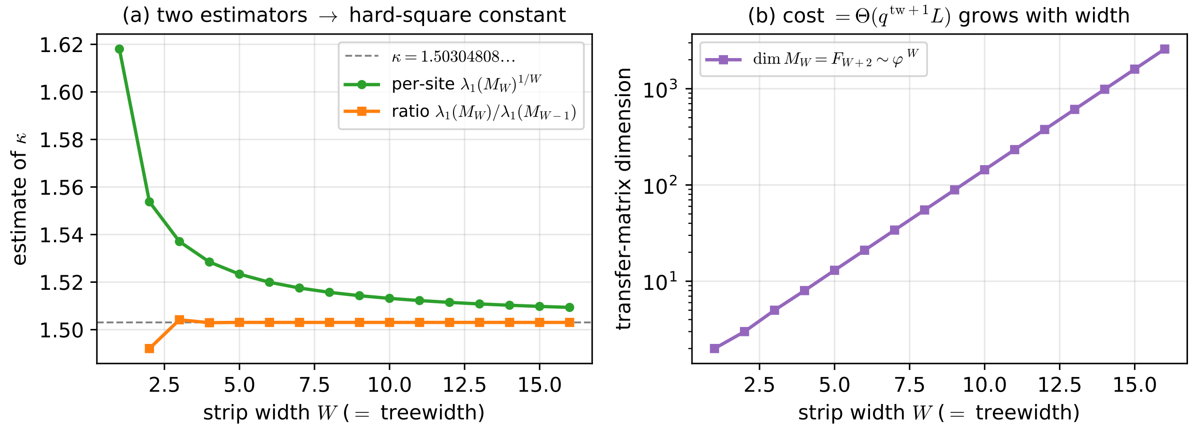

Figure 2: Hard squares on a width-W strip. (a) two estimators converge to κ (the ratio estimator geometrically); (b) the transfer-matrix dimension $F_{W+2} \sim \varphi^W$ — the cost — grows with the width.

3 Continued-fraction set E_2 — a transcendental dimension to 15 digits

Compute. The Hausdorff dimension of $E_2 = \{[0; a_1, a_2, \ldots] : a_i \in \{1, 2\}\}$, the reals whose continued-fraction digits are all 1 or 2. It is a transcendental constant $s^* = 0.5312805062772\ldots$[2] known only through such computations.

Series. For a word $I = (a_1, \ldots, a_L)$ let $\psi_a(x) = \frac{1}{a+x}$ and $\psi_I = \psi_{a_1} \circ \cdots \circ \psi_{a_L}$. The weight $\omega_I(s) = \sup_x |\psi_I'(x)|^s$ is multiplicative along the word (chain rule), and $Z_L(s) = \sum_{|I|=L} \omega_I(s)$.

[1] R. J. Baxter, *J. Phys. A* (1980); N. Calkin and H. Wilf, *SIAM J. Discrete Math.* (1998).

[2] O. Jenkinson and M. Pollicott, *Adv. Math.* (2001) and "*A hundred decimal digits...*" (2018), the reference value used here.

Engine & rate. The infinite Gauss map collapses onto a *fixed* operator acting on functions on $[0,1]$,

$$(\mathcal{L}_s f)(x) = \sum_{a\in\{1,2\}} (a+x)^{-2s}\, f\Big(\tfrac{1}{a+x}\Big),$$

whose leading eigenvalue $\lambda(s)$ is the rate of $Z_L(s)$. Bowen's equation gives the dimension as the root where the rate is 1:

$$\boxed{s^* \text{ is the unique } s \text{ with } \lambda(s)=1.}$$

We discretise $\mathcal{L}_s$ by Chebyshev collocation on $N{+}1$ nodes (*independent of word length*) and root-find in s.

Result. Two cross-checked routes (Figure 3):

- **operator:** at $N=16$ nodes, $s^*=0.5312805062772$ — correct to **15 digits**; convergence in N is spectral (each few nodes add several digits). The operator's gap is $\delta\approx 0.31$.
- **elementary ratio test:** enumerating all 2^L words, $Z_{L+1}(s)/Z_L(s)\to 1$ *exactly when* $s=s^*$: at s^* the ratio is 1.000000, while at $s^*\mp 0.03$ it locks onto 1.0387 and 0.9628. This is the textbook "set s to the guess, check the ratio $\to 1$" made rigorous.

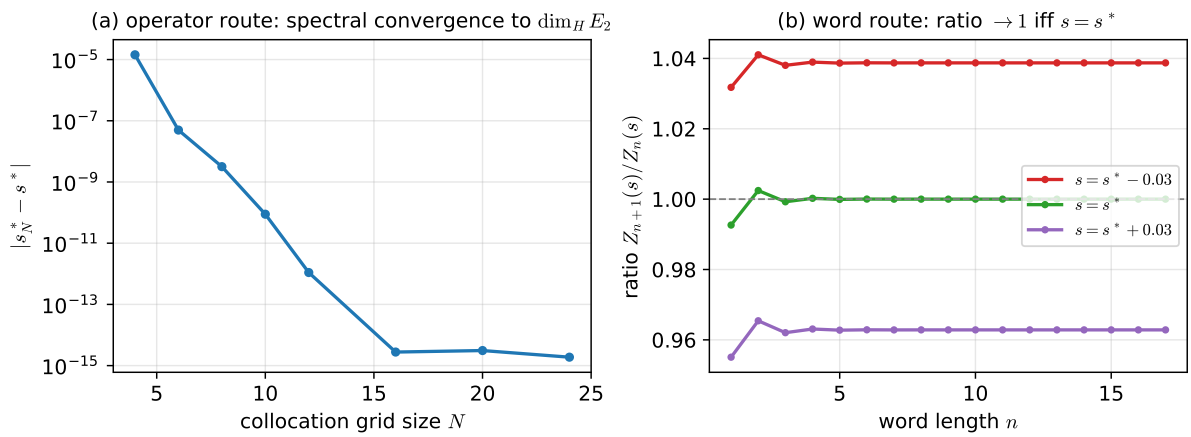

Figure 3: Dimension of E_2. (a) spectral convergence of the operator estimate; (b) the word-enumeration ratio tends to 1 only at the true dimension.

4 Matrix products — the entangled case, and when it collapses

Compute. Rates built from products $T_I = T_{i_1}T_{i_2}\cdots T_{i_L}$ of a fixed set of matrices: the joint spectral radius and Falconer's *affinity dimension*[3] of a self-affine set,

$$Z_L(s)=\sum_{|I|=L}\varphi^s(T_I), \qquad \varphi^s(A)=\begin{cases}\sigma_1(A)^s, & 0\le s\le 1,\\ \sigma_1(A)\,\sigma_2(A)^{s-1}, & 1\le s\le 2,\end{cases}$$

with $\sigma_1\ge\sigma_2$ the singular values. This weight is a *global* function of the *whole* product, so every index is coupled to every other: the dependency graph is the complete graph K_L, there is *no* finite transfer matrix, and exact evaluation costs q^L. This is the "everything in one lump" case. We use the explicit nonnegative pair

$$T_1=\begin{pmatrix}0.50 & 0.20\\ 0.10 & 0.40\end{pmatrix}, \qquad T_2=\begin{pmatrix}0.30 & 0.05\\ 0.25 & 0.45\end{pmatrix}.$$

[3]K. Falconer, *Math. Proc. Camb. Phil. Soc.* (1988).

(a) Collapse onto a decomposable surrogate. Replace the operator norm $\sigma_1 = \|\cdot\|$ by the *linear* functional tr. Linearity makes the weight multiplicative again, and the K_L entanglement collapses to a path:

$$\sum_{|I|=L} \operatorname{tr}(T_I) = \operatorname{tr}\big(S^L\big), \qquad S = T_1 + T_2, \qquad (\text{treewidth } 1,\ \text{cost } O(L)).$$

For *nonnegative* matrices Perron–Frobenius forces the norm-rate and this cheap surrogate to share one limit,

$$\boxed{\lim_{L\to\infty} \frac{1}{L} \log \sum_{|I|=L} \|T_I\| = \log \rho(T_1 + T_2).}$$

Numerically $\log \rho(S) = 0.114986731574$ (closed form), and the brute-force norm-sum over all 2^L words drifts onto it (Figure 4a). The honest boundary: for a *signed* pair the cancellation breaks the identity — our example gives norm-rate 0.149 against $\log \rho(S) = -0.153$ — and one then needs the projective transfer operator instead of the trace.

(b) Affinity dimension as a pressure root. The affinity dimension is the root of $P(s) = \lim_L \frac{1}{L} \log Z_L(s) = 0$. Brute force at $L = 18$ gives $s^* \approx 1.089$ (Figure 4b). The endpoint $s = 2$ is a *free check*: there $\varphi^2 = |\det|$ is fully multiplicative, so $Z_L(2) = (|\det T_1| + |\det T_2|)^L$ in closed form, and brute force reproduces $\frac{1}{L} \log Z_L(2) = -1.195674002$ to every printed digit.

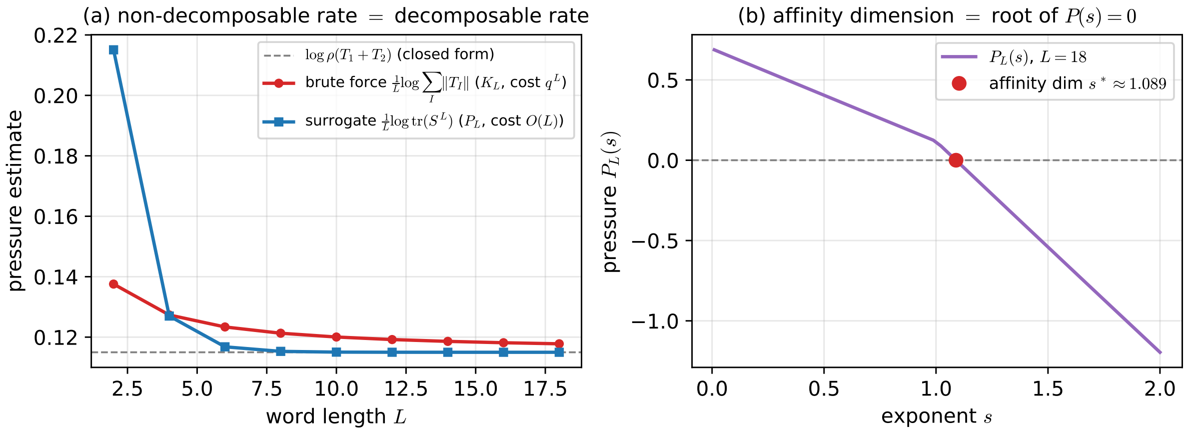

Figure 4: Matrix products (entangled, graph K_L). (a) the expensive norm-sum rate and the cheap trace surrogate share the limit $\log \rho(T_1+T_2)$ for nonnegative matrices; (b) the affinity dimension as the root of $P(s) = 0$, with $s = 2$ a closed-form validation.

Reproduce: `python3 code/run_all.py` regenerates every number and figure above; each script prints the reference constant it matches.